Metaphorical Usage as Contextual Embedding Shift: A Small Proof of Concept

Xianglong Liu
University of Tübingen

Abstract

This proposal reformulates a broad theory of figurative language as a focused, testable project on metaphor. The central question is whether metaphorical uses of a verb occur farther away from the verb's literal-use centroid in contextual embedding space. I operationalize $Enc(x, t)$ as the RoBERTa-base contextual embedding of target token t in sentence x, averaged over subwords when necessary, L2-normalized, and compared with cosine distance. A first MVP on VUA20 verb instances gives weak-to-moderate support for this hypothesis: metaphorical usages are significantly farther from lemma-specific literal centroids, but the effect is small to moderate, and simple classification is only slightly above the majority baseline. The next step is a polysemy control to test whether the observed shift is metaphor-specific or part of ordinary contextual sense variation.

1 Motivation and Scope

The original project asked whether figurative language can be modeled as a continuous semantic–pragmatic space rather than as a set of discrete rhetorical categories. This is theoretically attractive, but too broad for a controlled term project. I therefore reduce the scope to one device and one operational question: *can metaphorical verb usage be approximated as contextual embedding shift away from literal usage?*

This reduced project does not attempt to build a full metaphor detector or to prove a general theory of figurative language. It is a theory-driven proof of concept. The main contribution is to test whether a clearly defined contextual representation contains a measurable metaphor-relevant geometric signal within the same lemma.

The scope is deliberately narrow. I use English metaphor only, exclude simile, irony, sarcasm, idiom, humor, and discourse-level pragmatic inference, and use a real human-annotated corpus rather than LLM-generated examples. I also avoid the earlier abstract comparison between a literal meaning representation $Enc(S(x))$ and an intended meaning representation $Enc(I)$, because standard metaphor datasets usually provide sentences, target tokens, lemmas, POS tags, and literal/metaphorical labels, but not full intended meanings. The current project therefore measures a data-compatible proxy: the position of a target token in contextual embedding space.

2 Research Question and Hypotheses

RQ. Do metaphorical uses of a verb occur farther away from its literal-use centroid in contextual embedding space?

H1: Literal-centroid distance. For each lemma w, let the literal centroid be

$$c_{\mathrm{lit}}(w) = \frac{1}{N_w} \sum_{i=1}^{N_w} Enc(x_i, w), \quad (1)$$

where the sum ranges over literal usages of w. For each target usage, define

$$D_{\mathrm{lit}}(x, t) = 1 - \cos\big(Enc(x, t), c_{\mathrm{lit}}(t)\big). \quad (2)$$

The prediction is

$$D_{\mathrm{lit}}(x_{\mathrm{met}}, t) > D_{\mathrm{lit}}(x_{\mathrm{lit}}, t). \quad (3)$$

This is a weak geometric hypothesis: a positive result would suggest that metaphorical usage is associated with contextual shift, not that metaphor is reducible to cosine distance.

H2: Lemma-level heterogeneity. Some verbs should show clearer literal/metaphorical separation than others. This is expected because metaphorical extension is lexically uneven and may depend on conventionality, sense structure, and data sparsity. Lemma-level rankings are therefore exploratory unless each lemma has enough literal and metaphorical examples.

H3: Classification sanity check. If target-token embeddings contain metaphor-relevant signal, a simple classifier such as logistic regression or linear SVM should perform above a majority baseline. This is not intended as a state-of-the-art detection experiment.

H4: Polysemy control. The observed shift may reflect ordinary sense variation rather than metaphor-specific shift. A stronger analysis must compare metaphorical shift with literal–literal dispersion and, if possible, literal sense clusters:

$$D_{\text{same lit. sense}} < D_{\text{diff. lit. senses}} \leq D_{\text{lit}-\text{met}}. \quad (4)$$

If metaphorical distances are similar to ordinary polysemy distances, the result should be interpreted as evidence for contextual sense shift rather than specifically metaphorical geometry.

H5: Anisotropy robustness. Transformer embeddings are anisotropic, so raw cosine distance may reflect common directions or surface artifacts. The analysis therefore compares L2 normalization, mean-centering plus L2 normalization, and removal of the first principal component.

3 Method

3.1 Dataset

The MVP uses the VUA20 verb subset. This choice fits the project because it provides human metaphor annotation, target-level labels, and enough verb instances to support within-lemma comparison. I keep only lemmas with both literal and metaphorical instances, then sample a subset for a small, controlled MVP.

Item	Count
Total verb instances	24,930
Literal verb instances	18,703
Metaphorical verb instances	6,227
Eligible lemmas with both labels	784
Selected lemmas	200
Sampled rows	1,605
Literal sampled rows	898
Metaphorical sampled rows	707

Table 1: MVP v1.0 dataset statistics.

3.2 Encoding Function

In this project,

$$Enc(x, t) = \text{contextual embedding of target token } t \text{ in sentence } x. \quad (5)$$

The encoder is RoBERTa-base. If the target word is split into multiple subword tokens, I average their vectors. I then L2-normalize vectors before computing cosine distance. This makes $Enc()$ an explicit computational operation rather than an underspecified semantic function.

3.3 Experiments

Experiment 1: literal centroid distance. Compute each lemma's literal centroid and compare the distance distributions of literal and metaphorical usages. Report mean distance, distance gap, p-value, and Cohen's d.

Experiment 2: centroid separation by lemma. Compute the distance between literal and metaphorical centroids for each lemma. In the next version, apply a minimum threshold such as $n_{\text{lit}} \geq 5$ and $n_{\text{met}} \geq 5$ before interpreting lemma rankings.

Experiment 3: classification sanity check. Train logistic regression and linear SVM classifiers on target-token embeddings. Use grouped-by-lemma generalization to reduce lexical memorization. Report accuracy, macro-F1, metaphor F1, and the majority baseline.

Experiment 4: polysemy control. This has not yet been implemented. A minimal version computes leave-one-out literal–literal dispersion:

$$D_{\text{lit}-\text{lit}}(x, t) = 1 - \cos\big(Enc(x, t), c_{\text{lit},-x}(t)\big). \quad (6)$$

The diagnostic question is whether metaphorical distances exceed the upper tail of ordinary literal variation, for example the 75th or 90th percentile. A stronger version would cluster literal usages and compare literal-sense separation with literal–metaphor separation.

Experiment 5: anisotropy robustness. Compare three settings: L2 normalization, mean-centering plus L2 normalization, and removal of PC1 plus L2 normalization. The aim is geometric robustness, not necessarily better classification.

4 Current MVP Results

4.1 Distance Analysis

Experiment 1 supports the main hypothesis across all representation settings (Table **??**). Metaphorical usages are significantly farther from lemma-specific literal centroids than literal usages. However, the effect size is small to moderate, so the result should be framed as a geometric tendency rather than a full theory of metaphor.

Setting	Lit.	Met.	Gap	p	d
L2	.0549	.0621	.0072	$2.45e^{-7}$	.258
Mean+L2	.4508	.5260	.0751	$8.77e^{-15}$	.387
PC1+L2	.4600	.5331	.0731	$1.28e^{-13}$	.369

Table 2: Literal-centroid distance results. "Mean+L2" denotes mean-centering followed by L2 normalization; "PC1+L2" denotes removal of the first principal component followed by L2 normalization.

4.2 Classification Sanity Check

Classification is only slightly above the majority baseline of 0.556. The best logistic regression setting reaches 0.603 accuracy and 0.597 macro-F1. This suggests that RoBERTa target-token embeddings contain metaphor-relevant signal, but the signal is weak under grouped-by-lemma generalization.

4.3 Anisotropy Robustness

Raw RoBERTa embeddings are strongly anisotropic: the raw mean pairwise cosine is 0.8570 and the anisotropy score is 0.9285. Mean-centering and PC1 removal reduce common-direction effects and make the geometry more interpretable. However,

Setting	Model	Acc.	Macro-F1	Met.-F1
L2	LogReg	.603	.597	.558
L2	SVM	.595	.591	.555
Mean+L2	LogReg	.600	.595	.556
PC1+L2	LogReg	.597	.592	.551

Table 3: Classification sanity-check results. Majority baseline accuracy is 0.556.

they do not improve classification performance: removing PC1 changes macro-F1 by -0.0057 for logistic regression ($p = .3125$) and $+0.0038$ for SVM ($p = .8125$). Thus, anisotropy correction should be treated as a robustness and interpretability step, not as a performance-improvement method.

5 Expected Contribution and Next Step

The expected contribution is a small, transparent computational test of whether metaphorical verb usage corresponds to contextual embedding shift within the same lemma. The MVP already provides preliminary support for this weak hypothesis, but it also shows clear limitations: the effect is modest, classification is weak, and anisotropy correction changes geometry more than predictive performance.

The most important next step is the polysemy control. Without it, the project cannot determine whether metaphorical shift is distinct from ordinary polysemy or broader contextual sense variation. If metaphorical distances exceed literal–literal variation, the metaphor-specific interpretation becomes stronger. If not, the project still has a useful negative or mixed result: raw contextual embeddings may encode general contextual sense shift, while metaphor-specific interpretation requires additional semantic or cognitive supervision.

Limitations

The project uses only English verb metaphors, only one encoder in the MVP, and a sampled subset of VUA20. It does not model intended meaning, discourse context, audience interpretation, simile, or figurative language as a whole. The proposal therefore makes a limited claim: contextual embedding geometry may reflect a weak metaphor-relevant shift, but cannot by itself explain metaphor comprehension.

References

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